



CSC1108 BSc Data Science And Artificial Intelligence Formal Project Proposal

Project Title:	Predicting fourth quarter physical performance of elite male Gaelic footballers from GPS match data
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Executive Summary

Our project aims to develop a machine learning model to predict the fourth quarter physical performance of elite male Gaelic football players. Research has shown that a key differentiator of winning teams is their robustness to resist fatigue in the final quarter of games. By forecasting a player's running metrics, we can inform decision making by coaches and analysts regarding team selection and substitutions to maximise their teams output in these pivotal closing stages. Although there has been widespread descriptive analysis of GAA running statistics, there is no current implementation of machine learning technology to predict future outcomes. Our data comes from a publicly available dataset of one male inter-county team's running statistics over 3 seasons. Players movement data was recorded using wearable GPS tracking devices. Latitude, longitude and velocity readings have been converted into a temporal series of player actions to create a format suitable for machine learning. We will explore this data via statistical distributions and visualisations. This analysis will inform the necessary processing, transformations and feature engineering that we must perform. We will train a range of linear, ensemble tree based and deep learning models on data from historical and ongoing matches. We aim to achieve a high predictive accuracy whilst also understanding the importance of different features in training our models. Finally, we hope to produce a reusable pipeline for applying our methodology to other players and teams in the future.

Motivation & Background

Gaelic football is an Irish field invasion sport, governed by the GAA (Gaelic Athletic Association) [\[1\]](#). Matches (or games) are contested by two teams of 15 players on pitches around 135-140 metres in length and 80-90 metres in width. Players play for their local clubs at a grassroots level. Elite Gaelic football pertains to the top club players representing their counties in inter-county competition. Match duration at the elite level is 70 minutes split into 2 35 minute halves, plus additional time to account for injuries and other stoppages [\[2\]](#).

The physical demands for players within Gaelic football are constantly evolving due to changes in tactical strategies [\[3\]](#), increased athleticism [\[1\]](#) and adaptations to playing style [\[2\]](#). Structured defensive setups, advances in strength and conditioning and quicker transitional play have contributed to increased running loads across Gaelic football matches [\[4\]](#). Performance metrics such as total distance covered, high speed running distance and sprinting distance, [\[4\]](#) captured by Global Positioning System (GPS) data, have become pivotal indicators to determine the performance levels of players across Gaelic football matches. Whilst this data can be used to

determine player workload, it is generally used reactively to understand what occurred during a match, rather than predicting what will transpire next.

The motivation behind this project is to shift from descriptive analysis to focus on predictive modelling of the data produced by the GPS [4]. Our aim is to identify which players will perform best during the final quarter of a Gaelic football match. This focus is significant, as the final 15-20 minutes of a game are generally the most decisive periods, where fatigue and tactical decisions heavily influence the outcome of the match. This will allow coaches to make informed decisions [4] to identify underperforming players and manage workload [5] using expected output across different phases of a game. Machine learning methods could help in selective decisions, optimising substitutions and managing player activity before and after games.

Although extensive studies have analysed exertion levels [6], performance changes [7] and performance indicators from historic GPS data; they focus on what has already happened, not what will happen. Little work has been explored to predict the performance levels of players during the final quarter of a game to maximise a team's output. This indicates a current gap in the research behind GPS data within Gaelic Football.

Given that the GAA is Ireland's largest sports organisation [8], this presents an exciting opportunity to develop sports analytics further in this field. With over 2,200 clubs and 34 inter county teams [8], there is potential for the collection of vast amounts of player movement data and widespread application of our methodology, if proven to be effective. Player performance is often judged on basic analysis which is preventing teams from using players effectively. Progress in this area could demonstrate that future performance is not random and can be predicted from historic GPS data.

This problem is technically challenging due to the irregularity of the data logs produced by the GPS data [4]. Players tend to power on their GPS trackers sporadically, whether it's during the warm up, the beginning of the match or when the match has already commenced. Establishing stable patterns between players' data will require significant data handling, processing and feature engineering. Overcoming these challenges will allow us to identify which players were on the field-of-play at each moment. This research helps discover the physical indicators that have the most effect on the predictive power of the model. It emphasises the benefits of Gaelic football teams across the country collecting GPS data, thus advancing the role data science can play in GAA performance analysis.

Problem Statement

Our objective is to predict the performance of elite male Gaelic football players in the final quarter of games using GPS data. It has been shown that higher ranked teams have a higher running output and display greater physical intensity than lower ranked teams [9]. It is also true that a team's ability to maintain their running output across the four quarters is one of the key differentiators between winning and losing in Gaelic football [10]. We aim to determine whether an individual's running output will drop, remain stable or improve in the closing stages of a game.

We will leverage a database of in-game player movements from elite men's inter-county matches [11]. The data was generated using wearable GPS technologies, capturing features such as longitudinal and latitudinal positioning over time. Due to its high volume and granular nature, this data is typically unsuitable for machine learning [12]. In the case of [11], the raw data has already been transformed with this exact purpose in mind. Player movement has been categorised into 6 'action' labels, boundaried by velocity thresholds in m/s. This 6-zone system is a commonly applied framework for classifying movement data in Gaelic football [10], as well as sports such as soccer [13], Australian rules football [14] and hurling [15]. We will develop our model around historical movement patterns of individual players from previous matches, as well as load from ongoing games.

The scope of this work is to develop a predictive machine learning algorithm that will answer our 3 main research questions:

- To what extent can in-game GPS movement data be used to predict the running output of Gaelic football players in the fourth quarter of a match?
- Which velocity-banded action-type features are the strongest predictors of fourth-quarter running performance in Gaelic football?
- Are running metrics from previous full matches or from the first three quarters of the current match more predictive of fourth quarter running performance?

Our model will be able to inform real world decision making by managers, coaches and analysts. By accurately identifying fatigue-related performance decline in real time, they can optimise team selection and substitutions to maximise the physical output of their team. Teams that run further, faster and for longer in the most consequential, final period of the game have an increased probability of winning [10].

At inter-county level, there are no significant differences between the running profiles of players on different teams [16]. However, physical performance characteristics of club and county players are not statistically comparable [17]. The 6 zone velocity threshold framework is unsuitable for female sports as it overlooks the influence of

sexual dimorphism on pacing and recovery [18]. Applying cross-sport transfer learning has been found to reduce performance by 15-20% when training prediction models on player movement data [19]. This information suggests that although our approach will be validated in the specific context of one elite Gaelic football team, it will be highly generalizable to any men's senior inter-county team with access to player tracking data. However for female, cross-sport or grassroots implementation, our model risks low transferability without significant domain adaptation.

State of the Art

Research on physical performance in Gaelic football has grown significantly over the last number of years, mainly due to the increased availability of wearable GPS trackers during training and matches. GPS technology has become the primary method [20] for tracking players external load during field-based sports. Modern devices allow researchers to obtain performance indicators such as max acceleration, distance covered, high speed running load, sprinting frequency and sprinting location. However, before any research on meaningful analysis can occur, substantial preprocessing is required to ensure interpretation is possible. GPS data contains, as highlighted by Malone S. [4], high variability, irregular sampling intervals and high dimensional motion traces, which can cause irregular patterns to appear whilst working with the raw data.

As discussed previously, GPS units generate complex spatio-temporal datasets consisting of latitude, longitude and speed. The DetectAction [21] algorithm is used to sequentially process the raw data and compress consecutive similar movements into one single action, producing an event-level representation of player movements. The algorithm records start and end times, location and duration of the particular event. This transforms unprocessed high-frequency data into interpretable units of performance.

The GPS data is categorised into '6' predefined action labels, to form meaningful performance metrics. The standard framework [20] for Gaelic football is the 6-zone velocity threshold classification system. This system converts speeds into one of six action labels:

- 'standing' ($\leq 0.194\text{m/s}$)
- 'walking' ($0.194\text{m/s} < x \leq 2\text{m/s}$)
- 'jogging' ($2\text{m/s} < x \leq 4\text{m/s}$)
- 'running' ($4\text{m/s} < x \leq 5.5\text{m/s}$)
- 'high-intensity running' ($5.5\text{m/s} < x \leq 7\text{m/s}$)
- 'sprinting' ($x > 7\text{m/s}$)

This process ensures that movement events and workload profiling are built on behaviourally meaningful features.

The physical demands of Gaelic football players have been well documented. [20] acknowledges the requirement of players to cover significant distances across a variety of high intensity and unstable actions. Levels of physical demands are highly dependent on the player's position for the game. Running demands also vary significantly by position [21]. Players assigned to middle positions (half-back, midfield and half-forward) complete the highest distance-based measures, while full backs and full forwards accumulate smaller figures for these measures.

Running performance [4] of Gaelic football players changes as the match progresses. While direct longitudinal studies quantifying year-on-year increases in match-running outputs are scarce, several sources imply a trend of rising physical demands in Gaelic football. For instance, [22] shows improved conditioning capacities in youth players over a 10-year period, illustrating high running loads and fatigue associations that reflect increasing intensity demands. Previous samples taken during research have identified increased running performance from elite level players involved with inter-county teams. Collectively, these studies reinforce [23] the importance for elite Gaelic footballers to maintain high intensity running and sprinting consistently throughout a match to be able to compete at the highest level.

The Football Review Committee (FRC) introduced a suite of seven core rule enhancements [37] and other interconnected proposals that were trialled as temporary rules for the 2025 inter-county and club seasons. The goal of these enhancements was to reward skill, risk-taking, and forward momentum to create a more exciting and dynamic playing and spectator experience. Some of the modifications include a new 40 metre 2 point arc, kickouts beyond the 40 metre arc, the solo-and-go and the restrictions on the number of players in each half at all times.

Despite the introduction of these new rules into Gaelic football in 2025, the demands for high intensity running have not increased in the way some anticipated. Recent GPS analysis from FRC and the Games Intelligence Unit (GIU) suggests that the introduction of new rule enhancements hasn't produced a universal increase in physical demands. High-intensity running often decreased across most periods of play. While our data is gathered before these rule enhancements, the analysis only shows an increase in high intensity running during the final period of a game. This aligns directly with our project's focus on fourth-quarter performance, highlighting the relevance of our model despite the rule enhancements. Given the limited sample of

matches available in the 2025 analysis, the FRC also cautions that conclusions should be interpreted carefully.

Extensive research has marked [\[24\]\[10\]](#) the fourth quarter as a pivotal phase during a game which can merely determine the outcome of the match. Physical performance levels were generally maintained across all quarters in games that resulted in a win, in comparison to declines in performance observed in games lost. [\[4\]](#) analyses temporal patterns in match-play which shows that high-speed running distance declines progressively throughout quarters (second (-8.8%), third (-15.9%), and fourth (-19.8%)), with the steepest decline evidently occurring in the final quarter.

Fatigue [\[10\]](#) can be identified as a key contributor to these declines, as it is one of the effects of accumulated high intensity running during the first three quarters of a game. [\[4\]](#) demonstrates how an increase in fatigue can directly lead to a decrease in passing accuracy and defensive systems, two factors that deteriorate more sharply for losing teams during the final quarter.

Furthermore, the final quarter of games involves [\[23\]](#) a possession percentage decrease, increased transitional play and frequent momentum swings, which symbolise the value of tactical implications during this quarter. Since losing teams consistently demonstrate larger reductions in high-intensity running in the fourth quarter, understanding and predicting players performance during the final quarter is essential to maintaining tactical stability.

Player performance in Gaelic football can be defined by several dimensions. Technical skills such as shooting and passing accuracy, turnovers won and kickouts success are equally important as measures of physical output [\[23\]](#). Research shows a strong physiological link [\[24\]](#) between fatigue and technical decline. The gradual decrease of skill execution occurs simultaneously to reductions in physical performance. A strong physical performance influences the success of skill performance during a match, highlighting the importance of considering physical output during a game to understand a player's performance.

Analysis of winning patterns in elite Gaelic football matches demonstrates a sustained higher running performance during the final quarter of a match. Teams unable to match these demands decrease their probability of success in contested fixtures. An important way to maintain team performance in the later stages of a game is the use of substitutions. The introduction of substitutes into a game reported [\[25\]](#) a higher relative total distance, higher speed-distance and sprints per minute compared to

subbed off and full game players. The findings reflect the contribution of substitutions in elite-level Gaelic football.

These insights provide confidence in the relevance of modelling fourth-quarter performance for real-world decision-making.

Predicting the running performance of elite Gaelic footballers, in the last quarter of a match, will require machine learning capable of integrating real time GPS data and historical data. [6] has demonstrated how Gaelic football GPS data trained on machine learning models can successfully predict player RPE (ratings of perceived exertion). These results provide clear evidence that when Gaelic football GPS data is engineered appropriately, it exhibits predictable patterns suitable for predictive modelling. No published work has yet applied machine learning to predict a player's running performance during the final quarter of a game - a gap this review [21] identifies as having potential to optimise late game contributions and increasing the probability of winning.

Methodology

Our methodology combines best practices from sports analytics, machine learning and spatio-temporal data processing. The goal is to develop a predictive framework to estimate a player's expected movement patterns in the fourth quarter of a game using historical and real-time GPS data.

Previous work in Gaelic football has focused heavily on descriptive analysis. In particular, research has examined the relationship between running outputs and factors such as technical execution [26], positional differences [21], match outcomes [10] and team ratings [9]. Although there have been contradictory findings that both stronger [9] and weaker [16] teams cover more ground, studies consistently highlight an overall decrease in high-speed running and sprint distance across the course of games [29] [4]. It has been shown that teams who can withstand these declines and maintain their performance in the fourth quarter, greatly increase their chances of winning [10]. Despite these conclusions, there have been no solutions presented to forecast these late game activity levels.

Similarly, existing analytic tools such as the StatSports Sonra technology, most commonly used amongst elite GAA teams [28], only provide real time measurements and metrics. There is no capability to predict future performance.

It is in this way that our approach will stray from the current state of the art in Gaelic football research. We are not entirely reinventing the wheel but we aim to extend the existing literature in three key areas, each relating to our 3 main research questions.

- We will shift the focus from attempting to analyse and explain what has already happened in matches to predicting future events.
- We will investigate feature importance to determine exactly which running metrics are the most indicative of fourth quarter running output.
- We will model spatio-temporal data to extract deeper insights than summary level analysis provides.

Python will be the primary development language for our project. Its extensive libraries for machine learning, processing GPS data and time series analysis make it the most suitable choice for our project.

The main Python libraries we expect to employ are:

- Pandas / Numpy - Data processing and cleaning
- Matplotlib / seaborn - Visualisations and exploratory analysis
- Scikit-learn - Baseline classification and regression models
- XGBoost / LightGBM – Gradient boosting tree based models
- PyTorch / TensorFlow – Deep learning models
- TSFresh - Time series feature extraction

Our development environment will consist of:

- Jupyter Notebooks - Data processing, exploration, visualisations and early modelling
- VSCode - Code execution
- GitLab - Collaboration and version control

The primary dataset for our project is the database of in-game Gaelic football player movements referenced in [\[26\]](#). The data was collected from an unnamed men's inter-county team of 35 players during 11 matches across the 2019, 2020 and 2021 seasons. Players wore StatSports Gps trackers that provided readings of their latitude, longitude and speed (m/s) every 0.1 seconds. Movements of individuals are aggregated into their corresponding velocity-derived action types. Each row represents one event, a series of consecutive repeated actions, and specifies the event type,

duration, game and player. The exact gps coordinates have been removed to protect the anonymity of the team and players involved.

The database does provide a relatively small sample compared to the data used in current state of the art Gaelic football research [9][10][26][29]. However, due to a lack of a central GAA statistics repository and the reservedness of elite teams to share their data [3], this is the only publicly available dataset suitable for conducting machine learning analysis on player movement data from official games. As previously described, the data has already been collated and pre-processed with machine learning implementation in mind so no further collection and minimal transformation is required.

The main challenges connected to this data are the inconsistencies associated with GPS tracking data [27]. GPS logs can have irregular sampling rates. They may be switched on and off at imperfect intervals. They can also be subject to noise from signal interference. These factors can yield misaligned timelines between players that require careful cleaning and exploration before being usable.

Before we attempt to prepare our data for modelling, we will carry out a thorough explorative analysis of the raw GPS data as outlined in section 2.2 of [30]. Understanding the structure and statistical distributions of the data will allow us to identify outliers, missing values and timeline inconsistencies. We will then create visualisations to reveal performance patterns and trends within matches. Additional cleaning, transformations and feature engineering will be informed and driven by the findings of these investigations.

The main elements of our data processing will involve normalising the temporal elements within each game to a fixed interval to ensure all events are aligned on a consistent timeline. Variables such as additional time, substitutions, sending offs and human error in operating the GPS devices must be considered. We can then divide games up into accurate even intervals and aggregate player performance metrics across varying time windows. We will prioritise players who complete the entirety of each game but also explore the engineering of substitution-induced combinations. Two main categories of feature variables will be engineered for each player in each game:

- Historical features from previous matches (e.g. average distance covered per game)
- In-game features (e.g. current distance covered)

Our target variable ‘fourth quarter performance’ will be defined as one or more of:

- Total distance covered in the fourth quarter
- Total High Speed Running distance in the fourth quarter
- Total Sprint distance in the fourth quarter
- A categorical label (Increase/Decrease/Stable)

We will test multiple classification and regression models to determine the most effective predictor of final quarter running output.

Logistic and linear regression will be used as baseline models as they are highly interpretable and provide a benchmark for more complex algorithms to outperform [31]. Linear regression will predict the distances of each running type and logistic regression will estimate the probability of a player belonging to each of the three fatigue-profile classes.

Random forest [32] and gradient boosting [33] tree models are likely to demonstrate improved predictive accuracy. They can capture complex, non-linear interactions between the feature and target variables. Ensemble tree models have been favoured heavily in the sporting domain to extract spatiotemporal relationships from movement data [21]. These models are made more interpretable by the addition of SHAP values to explain their output [34].

We will also implement a Long Short Term Memory neural network. LTSMs are particularly strong at developing forecasting models using time-series data [35]. This is the most commonly applied deep learning approach for spatiotemporal sports performance analysis [36].

We will conduct 3 key experiments, each relating to one of our research questions.

- A baseline comparison of the performance of our linear, non-linear and deep learning models
- Using SHAP values to determine feature importance
- Feature ablation - Testing the model with historical features only, in-game features only and then combined

We will also carry out cross validation tests to ensure generalisation and prevent overfitting to specific players or matches.

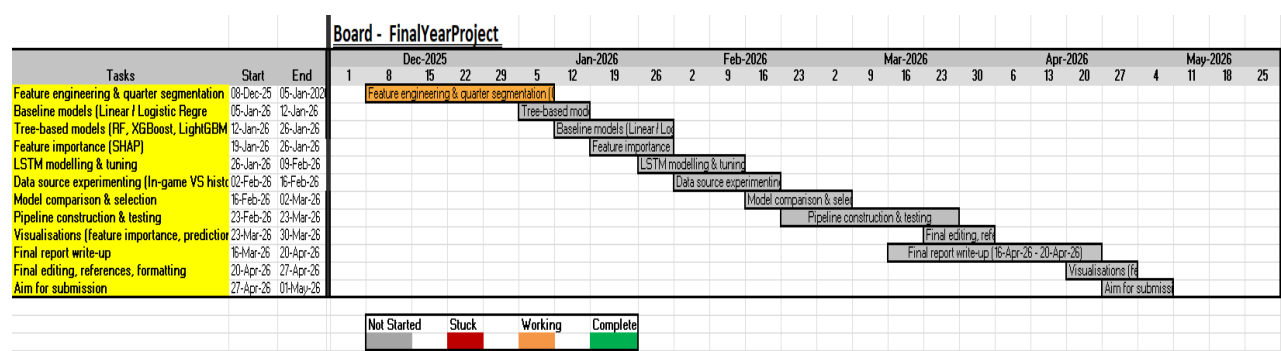
Our expectations for our project include training a strongly performing predictive model. Mean Average Error (MAE) for regression and F1 for classification will be the key metrics we use for evaluation. MAE fits our use case by providing coaches with a highly interpretable error margin on predicted distances that is not overly influenced by outliers. As we expect there to be less ‘increase’ labels than ‘stable’ and ‘decrease’, the F1 score will account for our imbalanced classes by combining precision and

recall values. We will examine secondary regression metrics of R^2 and mean absolute percentage error. Analysis of classification tasks will also consider overall accuracy and confusion matrices describing precision/recall per class.

Our other priorities include calculating a ranked list of our models most important predictive features to help coaches and players understand what causes fatigue and performance decline. We will create visualisations to display these feature importance distributions along with our predicted vs actual final quarter outputs. Finally, we will produce a reusable modelling pipeline for our approach to be applied to future movement datasets amongst elite Gaelic football athletes.

Gantt Chart

We have categorised our research into 9 main tasks to plan and manage our project workload. It is based on a realistic timeline of events from the conclusion of our first semester exams to the submission deadline at the end of semester 2. By producing a Gantt chart we can visualise our progress and clearly track deadlines and upcoming milestones. Our final drafting stage provides a reasonable buffer to allow for extenuating circumstances that may disrupt our timeline



Task 1 - Feature Engineering & Quarter Partitioning: Develop in-game elements, historical features, and precise fourth-quarter target factors. *(Weeks 1-5)*

Task 2 - Baseline Modelling (Logistic & Linear Regression): Determine interpretable baseline performance for both prediction and classification (increase, decrease or stable). *(Weeks 5-6)*

Task 3 - Tree-Based Modelling (XGBoost, Random Forest,): Train non-linear models capable of capturing interactions between historical and in-game outputs. *(Weeks 6-8)*

Task 4 - Feature importance (SHAP): To use SHAP values to identify the most influential features, which will help us comprehend fourth-quarter performance predictions. *(Weeks 7-8)*

Task 5 - LSTM modelling & tuning: Use sequential inputs to model player temporal behaviour across quarters for calculating Q4 load. *(Weeks 8-11)*

Task 6 - Data sourcing experimenting: Determine which features (historical vs current game vs both) have the most impact on prediction accuracy and evaluate how employing them affects performance. *(Weeks 11-13)*

Task 7 - Model selection and comparison: Determine the most effective method for forecasting fourth-quarter performance by comparing all trained models to standard measures. *(Weeks 12-17)*

Task 8 - Pipeline Construction & Final Evaluation: Build a usable pipeline for preprocessing, feature engineering, modelling, and prediction. *(Weeks 18-20)*

Task 9 - Final drafting: Finalise methods, visualisations, methods and critical evaluation into the final report. *(Weeks 17-22)*

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